Measuring matter

Matter takes up space. This means that it has **volume**. Volume is one measurement of matter. You can use a formula to calculate the volume of regular-shaped objects. You can also measure the volume of an object by dropping it in water and calculating the amount of water it **displaces**, or pushes aside.

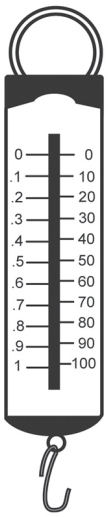
Mass is another measurement of matter. Mass describes the amount of matter or **atoms** in a substance. Matter can be measured by finding the **weight** of an object. An object's weight depends on its mass and how much **gravity** is pulling on it. For example, an object weighs less on the moon than on earth because there is less gravity on the moon. We can use a spring balance to measure weight.

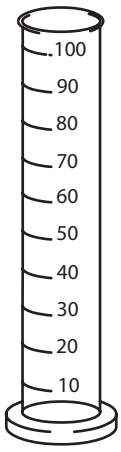
The **density** of an object depends on both its mass and its volume. Objects with tightly packed atoms are very dense. Objects with atoms spaced further apart are less dense. Density is measured in grams per cubic centimetre (g/cm^3) or kilograms per cubic metre (kg/m^3).

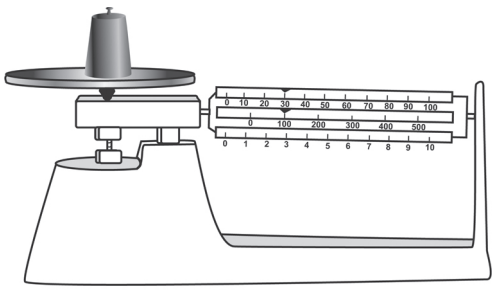
☞ **1** Read the information above. Underline the ways matter can be measured.

2 Many different instruments can be used to measure matter. Some are shown below. Use the words in the word bank to label each piece of equipment, then write down what each instrument could be used to measure.

Word bank
Spring balance
Triple beam balance
Measuring cylinder







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